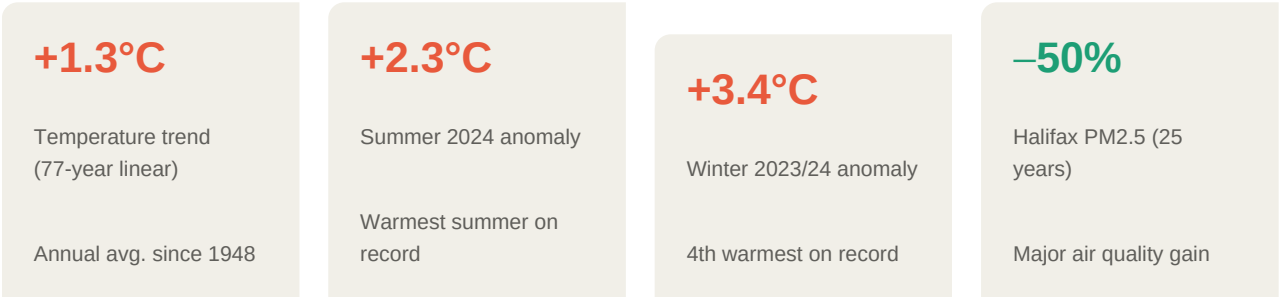


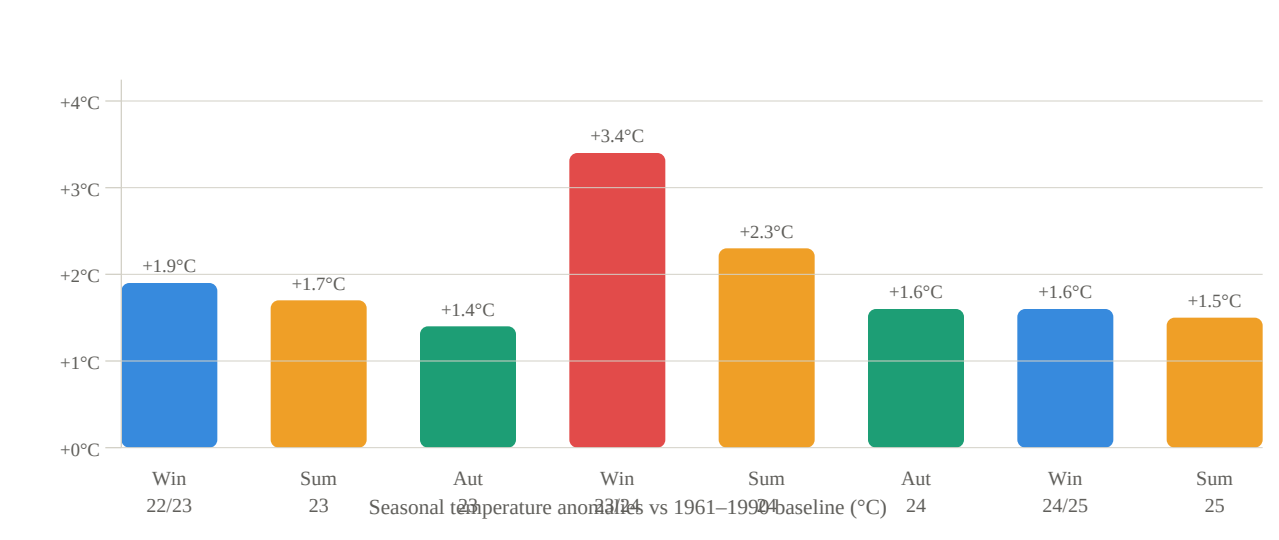
Atlantic Canada Climate & Air Quality

Precipitation, temperature, and pollutant trends · Data through 2025 · NS, NB, PEI, NL

KEY INDICATORS AT A GLANCE



TEMPERATURE ANOMALIES BY SEASON VS 1961–1990 BASELINE



▲ Warming trend

Consistent above-baseline warmth

Every season since 2022 has recorded above-average temperatures in Atlantic Canada. All 11 Canadian climate regions show positive long-term trends, with Atlantic Canada warming +1.3°C over 77 years — the most moderate gain nationally, yet still consequential for ecosystems.

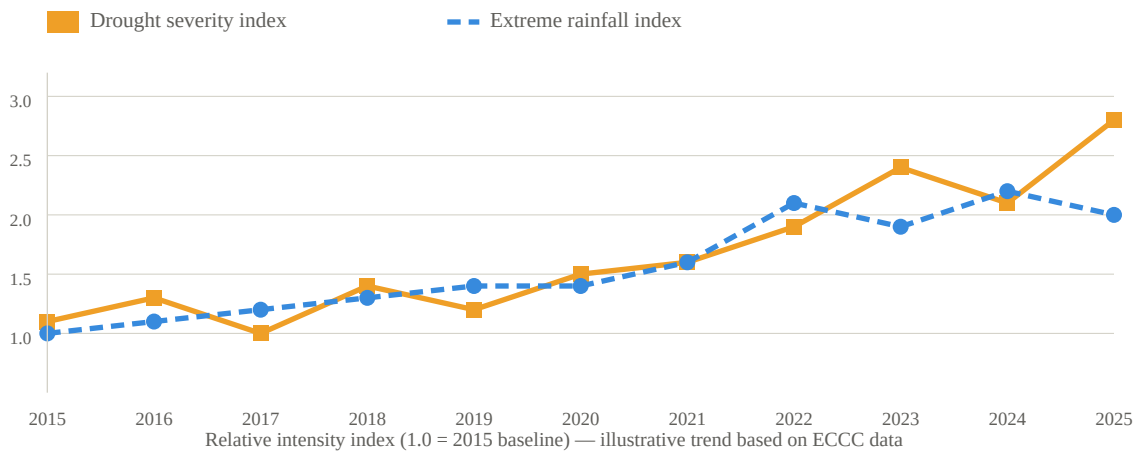
■ Extreme heat

2025 August heat wave

Atlantic Canada experienced the heat event with the strongest calculated human climate influence of any in summer 2025.

Newfoundland's Avalon Peninsula saw its longest-ever heat warning — 7 consecutive days, accompanied by drought and wildfire evacuations across the region.

PRECIPITATION & EXTREME WEATHER TRENDS



■ Drought-flood whiplash

Intensifying precipitation extremes

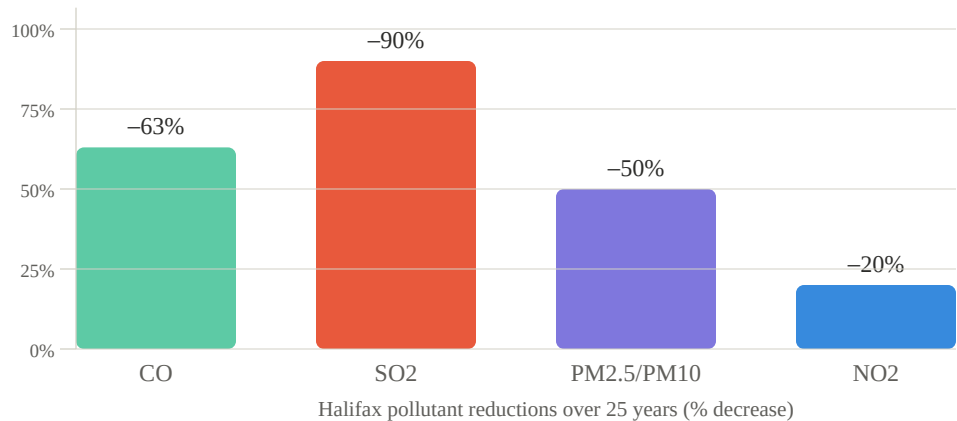
Atlantic Canada is experiencing prolonged dry spells punctuated by intense rainfall. Summer 2025 saw much of the Maritimes receive less than half their usual rainfall, pushing fire danger to extreme levels — while individual storm events deliver intense bursts of rain and coastal flooding.

■ Coastal flooding

Sea level rise & storm surge

The Atlantic coast faces sea level rise projections of up to 175 cm by 2100 under high-emission scenarios. Reduced sea ice cover is lengthening open-water seasons, increasing wave heights and storm surge intensity during winter. Hurricane Fiona (2022) demonstrated these compound flood risks on Sable Island and coastal communities.

AIR QUALITY & POLLUTANT TRENDS — HALIFAX (REPRESENTATIVE SITE)



✓ Long-term improvement

Steady reduction in industrial pollutants

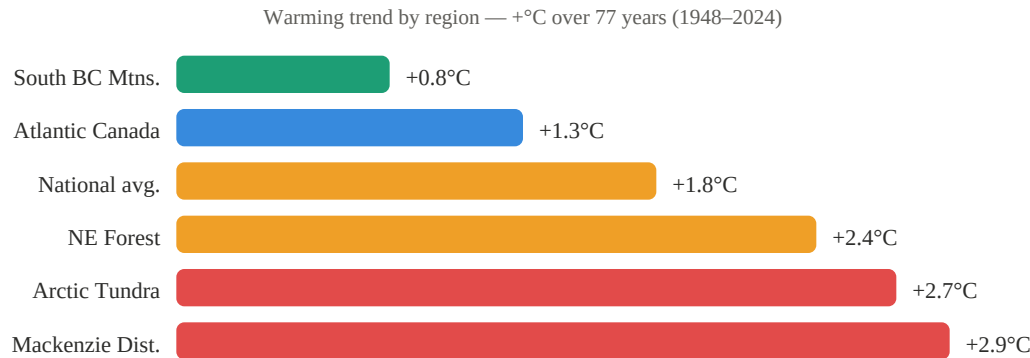
Atlantic Canada's relatively low population density keeps AQHI levels in the 1–3 (low risk) range most of the time. Decades of regulation have driven major reductions: SO₂ down 90%, CO down 63%, PM_{2.5} down 50% in Halifax. Transboundary ozone from the US northeast remains the dominant health-impact source.

■ Emerging threat

Wildfire smoke — a new risk vector

Wildfire smoke is an escalating threat. Between 2020–2024, smoke caused an estimated 1,400 deaths annually across Canada. Nova Scotia schools closed during the 2023 wildfire season due to hazardous air quality. The 2025 season ranked as Canada's second worst, with Atlantic provinces facing weeks of smoke-laden air.

TEMPERATURE WARMING RATE — ATLANTIC CANADA VS OTHER CANADIAN REGIONS



COMPOUNDING RISKS — HOW THESE TRENDS INTERACT

Warming → drought → wildfire smoke

Higher temperatures dry out forests; drought in 2025 drove extreme fire risk across the Maritimes, exposing residents to PM2.5 spikes that offset years of air quality gains.

Reduced sea ice → storm surge amplification

Loss of winter sea ice extends open-water seasons, increasing wave height and storm surge. Combined with sea level rise, coastal flooding risks in NS and NB are accelerating.

Warmer summers → ground-level ozone

Higher temperatures accelerate ozone formation from US transboundary precursors. Eastern Canada's ozone-attributable mortality is the dominant air pollution health impact in Atlantic provinces.

Sources: Environment and Climate Change Canada — Climate Trends & Variations Bulletins 2023–2025; Canada.ca air health trends; IQAir Halifax data; Canada-US Air Quality Agreement Review 2024; World Weather Attribution 2025; Climate Institute of Canada wildfire fact sheet 2025; NSCC Applied Geomatics Research Group; National Collaborating Centre for Environmental Health.